NovaTech Industries

Supplier Evaluation Business Rules — FR-005

Project: Intelligent Procurement & Logistics Management Platform Document ID: NVT-BRD-FR005 Version: 1.0 Status: Draft for business review Date: October 2026 Business owner: Procurement Manager Document classification: Internal — simulated project

Project note: NovaTech Industries, the supplier names, and the illustrative figures in this document are fictional. This document is part of a synthetic-data software development project.

1. Purpose and Scope

This document defines the supplier evaluation and recommendation requirements for FR-005. The platform shall assess supplier eligibility, compare eligible quotations using a weighted scoring model, explain the resulting ranking, and present a recommendation for human review.

Business objective: Improve purchasing decisions by balancing total acquisition cost, delivery performance, product quality, and supplier reliability.

In scope: Mandatory eligibility rules, scoring criteria and weights, quotation ranking, data inputs, recommendation output, traceability, and preliminary acceptance criteria.

Out of scope for this version: Automated purchase order approval, predictive demand planning, autonomous purchasing, and final normalization formulas (to be defined in Checkpoint 2.3).

2. Requirement Definition

Field	Definition
Requirement ID	FR-005
Requirement name	Weighted Supplier Evaluation and Recommendation
Business owner	Procurement Manager
Primary user	Procurement Officer
Requirement priority	Must
Decision authority	Procurement Officer / authorized approver

Requirement statement: The system shall evaluate eligible supplier quotations using a configurable weighted scoring model and recommend the most suitable purchasing option, subject to applicable business constraints and human approval.

2.1 End-to-End Decision Process

1. Eligibility assessment: Apply mandatory supplier and quotation constraints.
2. Weighted evaluation: Calculate normalized criterion scores for eligible quotations.
3. Ranking: Sort eligible quotations by final weighted score.
4. Recommendation: Present the highest-ranked quotation, its component scores, and an explanation.
5. Human approval: Require an authorized user to review the recommendation before a purchase order is issued.

3. Stage 1 — Mandatory Eligibility Rules

A quotation must satisfy all mandatory conditions before receiving a final evaluation score.

Rule ID	Mandatory condition	Expected system behavior
BR-001	Supplier has active status	Exclude inactive suppliers
BR-002	Supplier is approved for the product category	Exclude unapproved suppliers
BR-003	Quotation is valid on the evaluation date	Exclude expired or not-yet-valid quotations
BR-004	Supplier can fulfill requested quantity and minimum order conditions	Exclude infeasible quotations
BR-005	Quoted delivery meets required delivery date	Exclude quotations that cannot meet the mandatory deadline
BR-006	Supplier meets applicable quality and compliance requirements	Exclude noncompliant suppliers

Policy: An ineligible quotation cannot be recommended, even if it offers the lowest price. The system must record the specific exclusion reason(s).

4. Stage 2 — Weighted Supplier Evaluation

The initial evaluation policy uses four criteria. Weights are proposed configuration values and require business approval before production use.

Criterion	Code	Weight	Preferred direction
Total purchasing cost	C	40%	Lower is better
Delivery lead time	D	25%	Shorter is better
Product quality	Q	20%	Higher is better
Supplier reliability	R	15%	Higher is better
Total		100%	

Each criterion is normalized to a score from 0 to 100 before aggregation.

Weighted score formula:

$$S = 0.40 \times C + 0.25 \times D + 0.20 \times Q + 0.15 \times R$$

Where C, D, Q, and R are normalized criterion scores. The highest-scoring eligible quotation becomes the recommended option.

4.1 Configuration Controls

• Configured weights must sum to 100%; invalid configurations must not be activated.

• Weight changes must be versioned, dated, and attributable to an authorized user.

• Evaluation results must retain the weight configuration and underlying data used at the time of scoring.

• Tie-breaking rules will be defined before implementation; the system must not silently choose an arbitrary winner.

5. Illustrative Supplier Comparison

The following scores are already normalized. All three quotations are assumed eligible. They are examples only, not observed supplier performance.

Supplier	Cost C	Delivery D	Quality Q	Reliability R	Weighted score
EuroParts GmbH	80	95	92	96	88.55
Shenzhen Components	100	65	85	84	85.85
Nordic Industrial AB	72	90	98	95	85.15

Calculation example — EuroParts GmbH:

$$S = (0.40 \times 80) + (0.25 \times 95) + (0.20 \times 92) + (0.15 \times 96) = 88.55$$

Illustrative recommendation: EuroParts GmbH ranks first under the initial weight configuration. A lower-cost quotation does not necessarily produce the highest overall score because delivery, quality, and reliability also contribute.

6. Data Requirements

The following datasets and fields are needed to support FR-005. The final schema, keys, data types, and validation rules will be documented in the Data Dictionary.

Dataset	Minimum required fields	Purpose
suppliers	supplier_id, supplier_name, status, approval_status	Supplier eligibility
products	product_id, category_id, description	Product/category identification
supplier_products	supplier_id, product_id, minimum_order_quantity, available_quantity	Supply feasibility
purchase_requisitions	requisition_id, required_delivery_date, evaluation_currency	Purchasing constraints
requisition_items	requisition_id, product_id, requested_quantity	Requested products and volumes
quotations	quotation_id, supplier_id, valid_from, valid_until, currency	Quotation eligibility and currency
quotation_items	quotation_id, product_id, quoted_quantity, unit_price, lead_time_days	Commercial comparison
supplier_performance	supplier_id, evaluation_period, quality_score, on_time_delivery_rate, sample_size	Historical performance
evaluation_criteria	criteria_version, criterion_code, weight, normalization_method	Scoring configuration
supplier_evaluations	evaluation_id, quotation_id, criteria_version, component_scores, final_score, evaluation_timestamp, eligibility_result	Auditability and results

Additional data considerations: Freight costs, duties, exchange rates and their effective dates, compliance requirements, quotation-specific delivery dates, and approval records may require separate fields or tables. The data dictionary will resolve these details.

7. Business Policy Considerations

7.1 Total Purchasing Cost

Comparisons should use a consistent total purchasing cost rather than unit price alone. Depending on purchasing terms and data availability, this may include merchandise cost, freight, duties, and other directly attributable acquisition charges. The precise cost boundary must be documented.

7.2 Currency Conversion

Quotations in different currencies must be converted to a common evaluation currency using a recorded exchange rate, rate date, and source. Comparisons must not mix raw amounts denominated in different currencies.

7.3 Score Normalization

Raw measures (such as EUR and days) must be transformed into comparable scores between 0 and 100. The normalization methodology, treatment of outliers, and sensitivity to the set of quotations will be specified in Checkpoint 2.3.

7.4 Historical Supplier Performance

Quality and reliability scores must be derived from defined performance measures and an explicit observation period. Missing or insufficient history requires a documented policy; the system must not assume an unverified perfect score.

7.5 Explainability and Human Oversight

The recommendation interface must display eligibility decisions, criterion scores, configured weights, final rankings, and supporting data. Recommendation does not constitute purchasing approval.

8. Functional Acceptance Criteria

FR-005 is ready for acceptance testing when the system can demonstrate that it:

1. Excludes inactive suppliers.
2. Excludes suppliers not approved for the relevant product category.
3. Excludes quotations that are not valid on the evaluation date.
4. Rejects quotations that do not satisfy quantity and minimum-order constraints.
5. Enforces mandatory delivery-date constraints.
6. Applies applicable quality and compliance requirements.
7. Calculates comparable purchasing costs in the selected evaluation currency.
8. Normalizes the four evaluation criteria according to the approved method.
9. Validates that active weights total exactly 100%.
10. Ranks eligible quotations by weighted score.
11. Explains the recommended supplier and the reasons for exclusions.
12. Handles no-eligible-supplier and incomplete-data cases without issuing a recommendation.
13. Preserves an auditable record of inputs, scoring configuration, results, and reviewer decision.
14. Requires authorized human review before a purchase order is issued.

9. Open Decisions and Next Checkpoint

Decision	Status	Next action
Evaluation strategy	Approved for project design: weighted score	Implement after data model is finalized
Initial weights (40/25/20/15)	Proposed	Validate with procurement stakeholders
Normalization formulas	Open	Define in Checkpoint 2.3
Missing-data policy	Open	Define in Checkpoint 2.3
Tie-breaking logic	Open	Define before implementation
Cost inclusion rules	Open	Define before database implementation
Performance observation period	Open	Define before data generation

Next checkpoint: 2.3 — Score Normalization & Data Validation. Define the conversion of raw business measures to normalized scores, input validation, missing-data policies, and edge-case handling. The approved outcomes will then be incorporated into the broader Business Requirements Document (docs/business/business-requirements.md).